

EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

REVISION

GS-1 (DYNAMIC)

MODULE - 24



**ECOLOGY
&
ENVIRONMENT-5**

What is the present status of India as far as protected areas are concerned?



- MoEFCC is mandated to protect and improve the environment and safeguard the forest and wildlife of the country under Article 48-A of the Constitution.
- The number of protected areas in the country are now 1,022, accounting for 5.43% of the country's total geographic area.
- The number of Community Reserves are now 220. India has 58 tiger reserves and 33 elephant reserves.
- At present, India has 96 Ramsar Sites covering an area of 1.4 million hectares. India boasts the largest Ramsar site network in Asia and the 3rd largest in the world in terms of number of sites.
- India hosts 70% of the wild tiger population in the world and the number stands at 3,682, as per All India Tiger Estimation 2022.



- On the advice of National Tiger Conservation Authority (NTCA), three tiger reserves were notified under Section 38(5) of the Wildlife Protection Act, 1972.
- The number of tiger reserves in the country have gone to 58.
 - ✓ Guru Ghasidas-Tamo Pingla Tiger Reserve with core area 2,049 sq kms and total area 2,829 sq kms.
 - ✓ Ratapani Tiger reserve with core area 763 sq kms and total area 1,271 sq kms.
 - ✓ Madhav Tiger Reserve, Madhya Pradesh.
- The area under the tiger reserve network is now 82,836 sq km, which is roughly 2.5% of the total geographical area of the country.

On March 9, 2025 the Union Environment Minister announced that Madhav National Park in Madhya Pradesh is the country's 58th Tiger Reserve. This is the 9th Tiger Reserve in the state, the highest among the states. Maharashtra has 6. Rajasthan, Tamil Nadu and Karnataka have 5 each.

Project Tiger was replaced by the National Tiger Conservation Authority (NTCA) since 2006.



TIGERS WERE AT AROUND 40,000 AT THE TURN OF THE 20th CENTURY

- In 1964, it was estimated that there would have been around 40,000 tigers in the country at the turn of the 20th Century.
- By the 1960s, these numbers were down to between 2,000 and 4,000. The reasons are attributed to wanton hunting aided by a proliferation of gun licences issued in the years following 1947, improved access to the forest, clearing of large tracts of forests for various purposes, mushrooming of the new businesses of “Shikar Companies” and fur trade.

HOW WAS THE PROJECT TIGER EVOLVED IN 1970s?



- The origin goes back to 1969. The then Indian Board for Wild Life (IBWL), in a meeting in New Delhi in July, 1969 recommended a total ban on the export of all wild cat skins, including tigers.
- The same year, the 10th Assembly of IUCN met in Delhi and included the tiger in its “Red Data Book” as an endangered species and adopted a resolution calling for a ban on the killing of tigers.
- As the numbers dwindled to 1,863, the then Prime Minister commissioned a 11-member Task Force to investigate the problem and prepare a project report to preserve the tiger in the wild in India.
- In August 1972, the Task Force recommended eight tiger forests spread across India to be brought under the purview of “Project Tiger”.
- On April 1, 1973, Project Tiger was inaugurated at Corbett tiger reserve with nine tiger reserves announced across India.

THIS SLIDE GIVES IDEA ABOUT TIGER RESERVES IN INDIA

Homes for the big cat



Tiger reserves are spread across the length and breadth of India, with Madhya Pradesh having the highest number of nine reserves

As of the 2023 tiger census, India is estimated to have 3,682 tigers. About 30% of them are considered to be living outside tiger reserves



- As per 2023 tiger census, India is estimated to have 3,682 tigers. About 30% of them are considered to be living outside tiger reserves.
- Madhya Pradesh has the highest number of tigers at 785. It boasts several prominent tiger reserves, such as Kanha, Panna, and Bandhavgarh.
- However, the Kuno-Madhav forest division, in the northern part of the State, has historically been relatively neglected. Now, Kuno got prominence as an emerging cheetah reserve.

HOW ARE THESE TIGER RESERVES FUNDED?



- Under Project Tiger guidelines, 60% of the funds for conservation are provided by the Centre, while the rest is borne by the concerned State. In the case of Northeastern and Himalayan States, it is 90:10.
- The activities for which this money can be used include:
 - ✓ Anti-poaching initiatives.
 - ✓ Habitat improvement and water development.
 - ✓ Addressing human-animal conflicts.
 - ✓ Designating inviolate spaces.
 - ✓ Relocating villages from critical tiger habitats within a timeframe by offering a better relocation package.
- Project Tiger also supports States in settling the rights of displaced people, rehabilitating traditional hunting tribes living in and around tiger reserves, conducting independent monitoring, and evaluating tiger reserves.

WHAT IS THE PROCEDURE FOR ESTABLISHING A RESERVE?

To establish a reserve, the Centre gets a proposal from the State, the NTCA recommends the proposal to the State after due diligence, the State government notifies the area as a Tiger Reserve.



WHY IS MADHAV NATIONAL PARK IMPORTANT?

- It did not have tiger population till 2023, when a male tiger and two females were relocated there. Today, the population has grown to seven.
- However, the Madhav Tiger Reserve is an important connecting corridor to the Ranthambore Tiger Reserve in Rajasthan. It is also connected to Kuno National Park, which now has a captive cheetah population.



- Pathogen Access and Benefit-Sharing System (PABS), is designed to facilitate the swift sharing of pathogen data among nations and with pharmaceutical companies. This is expected to accelerate the development of vaccines, diagnostics, and treatments during future pandemics.
- Pharmaceutical companies have committed to donate 10% of their production to WHO and offer up to another 10% at affordable prices.
- Countries have agreed that technology transfer will be on mutually agreed terms and not voluntary. This is to balance the needs of global equity with the business realities of pharmaceutical innovation.



RECENT SUPREME COURT JUDGMENT ON SACRED GROVES

- The Supreme Court has directed the Forest Department of Rajasthan to identify and notify sacred groves in the state with detailed on-ground and satellite mapping.
- The Court has also recommended that the Union Minister of Environment create a comprehensive policy for the governance and management of sacred groves across the country.
- As a part of this policy, the MoEFCC must also develop a plan for nationwide survey of sacred groves, by whatever name they are identified in each state.
- This survey should identify the area, location and extent and clearly mark their boundaries.

Sacred groves are patches of land that are dedicated to local deities or ancestral spirits and are often preserved with near natural vegetation.

WHAT DO YOU UNDERSTAND BY SACRED GROVES?

Learning Space



- Sacred groves are community-regulated and conserved patches of forest land. They are created and managed traditionally by communities in various ways deeply rooted in their identity.
- These groves are kept inviolate through customary laws and taboos.
- The grove's conservation ethos is strongly linked to piety and sanctity, enforced in terms of the community's relationship with its spirits and gods.
- The sacred groves of Rajasthan, also known as 'orans', 'malvan', 'deo ghat', and 'baugh', number around 25,000 and cover about six lakh hectares of the State.
- They are called 'devara kadu' in Karnataka, 'devban' in Himachal Pradesh, 'kavu' and 'sarpa kavu' in Kerala.

SUPREME COURT ALSO CITED PROVISIONS OF THE FOREST RIGHTS ACT, 2006



Forest Rights Act 2006

- Citing Forest Rights Act, 2006, which recognizes the rights of tribal communities on forest resources, the apex court directed Rajasthan to identify traditional communities that have historically protected sacred groves and designate the areas as community forest resources under FRA.
- These communities have shown a strong cultural and ecological commitment to conservation and their role as custodians should be formally recognized, as per the judgment.
- As per Section 5 of the Forest Rights Act, they should also be empowered along with Gram Sabhas and local institutions, to continue protecting wildlife, biodiversity and natural resources. Granting authority to regulate access and prevent harmful activities would preserve their legacy of stewardship, as per the judgement.

THE SUPREME COURT JUDGMENT IS IMPORTANT FROM TWO ASPECTS

LEGAL PROTECTION

- It gives legal protection to the sacred groves and brings them under the purview of Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980.
- This implies that its non-forest use would require prior permission from the Union government.



IMPORTANCE OF COMMUNITY PROTECTED FORESTS

It highlights the importance of community protected forests, which would encourage local communities to not only preserve the Orans but also be an important part of the decision-making process at the state and central levels.

The judgment also upholds the customary rights of the communities and recommends the Orans to be declared as community reserves under the Wildlife Protection Act, 1972.

What is the historical perspective of the Intergovernmental Negotiating Committee (INC) to address plastic pollution?



- The fifth session of the United Nations Environment Assembly (UNEA) in 2022 adopted a historic resolution to address plastic pollution at the global level.
- The resolution mandated the Intergovernmental Negotiating Committee (INC) to develop an international legally binding instrument on plastic pollution and set into motion, a process to have the global agreement.
- The UNEA resolution took an ambition to conclude the negotiations of the INC by 2024. Four sessions of the INC have been held since 2022 in Uruguay, France, Canada and Kenya.
- The fifth session of the INC was held recently in Busan, South Korea. It is the last planned session of the INC but there was no breakthrough.

WHAT IS THE TECHNOLOGY USED FOR GENOME EDITED RICE VARIETIES?



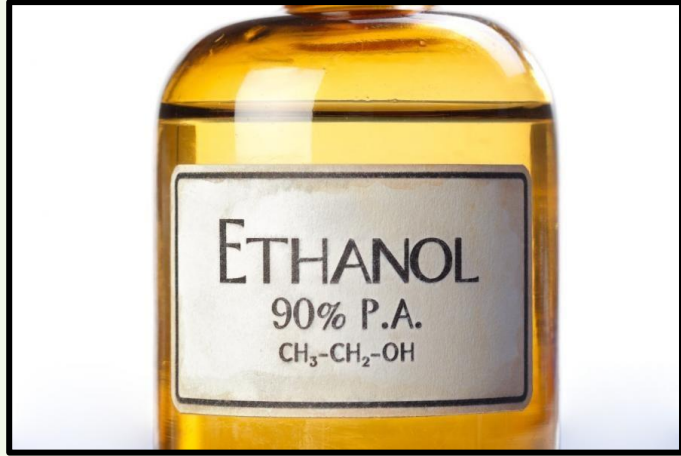
- Scientists have used Site-Directed Nuclease 1 and Site-Directed Nuclease 2 (SDN-1 and SDN-2) genome editing techniques to develop the seeds.
- Though this technique was used to develop different crops since 2001, such as tomatoes, a fish variety in Japan and a soybean variety in the U.S., making a rice variety has been done for the first time.

The International Research Community has approved both the varieties, as per the government.

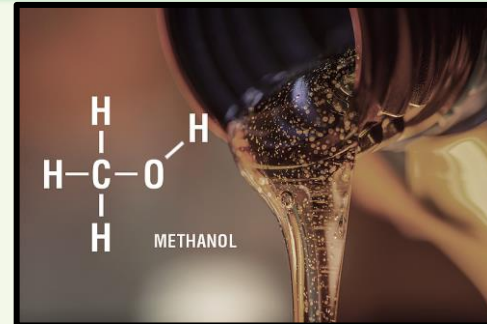
ARE THEY CONSIDERED GENETICALLY MODIFIED CROPS?



- They are not considered Genetically Modified (GM) crops as SDN-3 is not involved in this process.
- In the SDN-1 approach, scientists make a cut and the repair is done automatically, while in SDN-2, scientists give guidance to the cell to do the repair and the cell copies it.
- In this case, the mutant was developed without any foreign gene and mutation occurred through natural process. This is a precision mutation technique and several countries have exempted this process from the regulations required for developing GM crops.
- In SDN-3, scientists introduce a foreign gene from other varieties and integrated into the improved varieties. This process is considered as genetic modification.

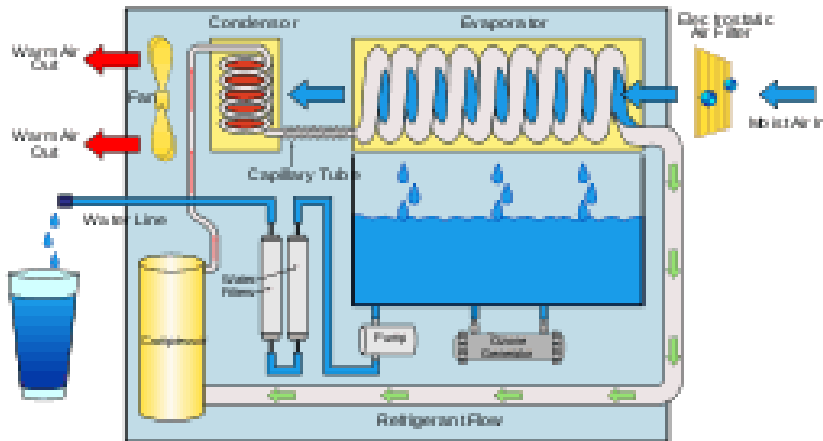


- Ethanol has two carbon atoms and Methanol one carbon atom.
- The chemical composition of Ethanol is $\text{CH}_3\text{CH}_2\text{OH}$, whereas for methanol it is CH_3OH .
- Methanol is the low Carbon, Hydrogen carrier fuel from high Ash Coal, Agri-residue, Carbon Dioxide from thermal power plants.
- Coal Reserves which are difficult to tap, and Municipal Solid Waste can be converted into Methanol. It is slightly low in energy content. It can be used in transport/energy sectors.
- It is poisonous.
- Dimethyl Ether (DME) can be mixed in LPG up to 20%.





- Tamil Nadu banned the sale of cotton candy after confirming the presence of Rhodamine B.
- Karnataka Government has banned the use of harmful colouring agents in cotton candy and Gobi Manchurian.
- Rhodamine B is a textile dye and its use in food has a huge impact on health. It is a fluorescent dye used in cosmetics, textile and leather industries. It gives brilliant pinks, greens and blues.
- Unfortunately, it is used as a food colouring agent, not only in cotton candy, but also in the preparation of sweets, various manchurian items and pakodas.
- Studies have shown that Rhodamine B can cause cell death. In long-term use, it can damage the cerebellum tissue and brain stem. There is recent evidence that it damages the kidney, liver and increases the risk of stomach tumour.
- It is toxic to the human body and is a carcinogen, as per experts.



- The purpose of atmospheric water generators is to harvest water from thin air. They are emerging as a potential solution for regions facing water scarcity or contamination.
- There are two main methods.
 - ✓ Similar to Air Conditioners, warm air is passed over chilled coils. As the air cools, moisture condenses and is collected, filtered and stored. This is energy intensive and is suitable for homes due to its consistent water production. This is known as condensation method.
 - ✓ The second method is desiccation. This uses absorbent materials (desiccants) to capture moisture from the air. The saturated desiccants are then heated, releasing water vapour which is condensed and filtered. This approach is more energy efficient and often used in larger settings.

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Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

98499 42299